1. We consider the given polyhedron and the feasible basic solution with x_2 and x_4 in the basis. The corresponding feasible basic solution is

$$x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \end{pmatrix}$$

and the permutation of the columns to have all basic variables first is defined by the matrix

$$P = \left(\begin{array}{cc|cc} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{array}\right).$$

The basic direction corresponding to the non basic variable x_1 is

$$d_1 = P \begin{pmatrix} -B^{-1}A_1 \\ 1 \\ 0 \end{pmatrix} = P \begin{pmatrix} -\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ 1 \\ 0 \end{pmatrix} = P \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \\ -2 \end{pmatrix},$$

and the basic direction corresponding to the non basic variable x_3 is

$$d_3 = P \begin{pmatrix} -B^{-1}A_3 \\ 0 \\ 1 \end{pmatrix} = P \begin{pmatrix} -\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \\ 1 \end{pmatrix} = P \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \\ -1 \end{pmatrix}.$$

Moving along d_1 with a step $\alpha \geq 0$, we obtain the point

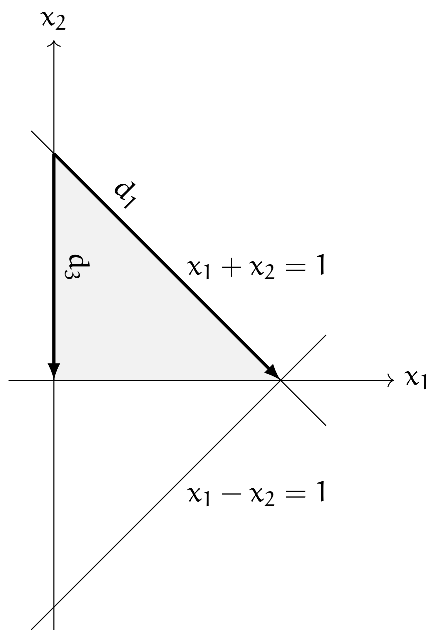
$$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ -1 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} \alpha \\ 1-\alpha \\ 0 \\ 2-2\alpha \end{pmatrix},$$

that is feasible for each $0 \leq \alpha \leq 1$. Therefore, the direction is feasible. Moving along d_3 with a step $\alpha \geq 0$, we obtain the point

$$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ -1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1-\alpha \\ \alpha \\ 2-\alpha \end{pmatrix}$$

that is feasible for each $0 \leq \alpha \leq 1$. Therefore, the direction is also feasible.

These two directions are shown in the following figure at point $(0, 1)$, where their length corresponds to $\alpha = 1$.



2. We now consider the feasible basic solution where x_1 and x_4 are in the basis. The feasible basic solution is

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

and the permutation matrix is

$$P = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{array}\right).$$

The basic direction corresponding to the non basic variable x_2 is

$$\tilde{d}_2 = P \begin{pmatrix} -B^{-1}A_2 \\ 1 \\ 0 \end{pmatrix} = P \begin{pmatrix} -\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ 1 \\ 0 \end{pmatrix} = P \begin{pmatrix} -1 \\ 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 2 \end{pmatrix},$$

and the basic direction corresponding to the non basic variable x_3 is

$$\tilde{d}_3 = P \begin{pmatrix} -B^{-1}A_3 \\ 0 \\ 1 \end{pmatrix} = P \begin{pmatrix} -\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \\ 1 \end{pmatrix} = P \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

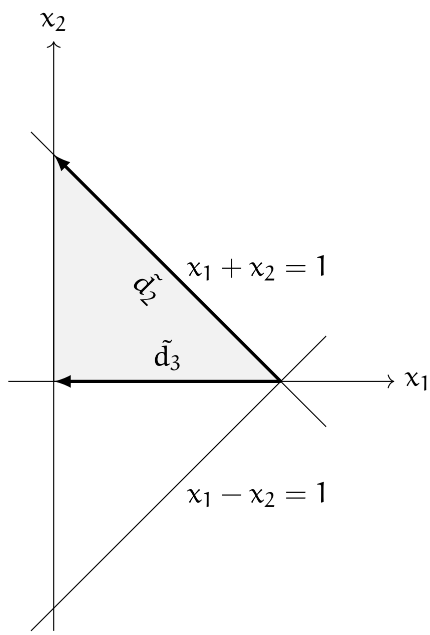
Moving along $\tilde{d}_2$ with a step $\alpha \geq 0$, we obtain the point

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} -1 \\ 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1-\alpha \\ \alpha \\ 0 \\ 2\alpha \end{pmatrix},$$

that is feasible for any $0 \leq \alpha \leq 1$. Therefore, the direction is feasible. Moving along $\tilde{d}_3$ with a step $\alpha \geq 0$, we obtain the point

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1-\alpha \\ 0 \\ \alpha \\ \alpha \end{pmatrix},$$

that is feasible for any $0 \leq \alpha \leq 1$. Therefore, the direction is feasible. These two directions are shown in the following figure at point $(1, 0)$, where their length corresponds to $\alpha = 1$.



3. Finally, consider the feasible basic solution where x_1 and x_2 are in the basis. Then,

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \text{ and } P = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) = I.$$

The basic direction corresponding to the non basic variable x_3 is

$$\hat{d}_3 = \begin{pmatrix} -B^{-1}A_3 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \\ 0 \end{pmatrix},$$

and the basic direction corresponding to the non basic variable x_4 is

$$\hat{d}_4 = \begin{pmatrix} -B^{-1}A_4 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 0 \\ 1 \end{pmatrix}.$$

Moving along $\hat{d}_3$ with a step $\alpha \geq 0$, we obtain the point

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{2}\alpha \\ -\frac{1}{2}\alpha \\ \alpha \\ 0 \end{pmatrix},$$

that is not feasible for any $\alpha > 0$ as the second entry is negative. Therefore, the direction is **not** feasible.

Moving along $\hat{d}_4$ with a step $\alpha \geq 0$, we obtain the point

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{2}\alpha \\ \frac{1}{2}\alpha \\ 0 \\ \alpha \end{pmatrix},$$

that is feasible for any $0 \leq \alpha \leq 2$. Therefore, the direction is feasible.

These two directions are shown in the following figure at point $(1, 0)$, where their length corresponds to $\alpha = 1$. Note that $\hat{d}_4$ is a feasible direction, whereas $\hat{d}_3$ is not.